INGÉNIEUX

IEEE ANNUAL EVENT

12-HOUR HACKATHON CHALLENGE

Forests & Wildlife Intelligent Sensing

4 NOVEMBER 2026 | MAHINDRA UNIVERSITY | TEAMS OF 1-4 | PRIZE POOL: INR 12,000

THE THEME

Forests and wildlife ecosystems generate diverse signals that reveal animal activity, habitat conditions and potential threats. Turning these observations into reliable insights is challenging because signals may be noisy, incomplete or affected by changing environmental conditions.

THE CHALLENGE

Develop an intelligent system that addresses a meaningful problem within the forests and wildlife domain. Teams must formulate their own problem and integrate Signal Processing with Agentic AI. The signals must be meaningfully analysed, and the intelligent component must use its observations to reason, make context-aware decisions, adapt or take appropriate actions.

WHAT THE SOLUTION SHOULD DEMONSTRATE

AGENTIC AI

The system must go beyond a one-time classification or fixed prediction. Its agentic component should reason about observations, respond to context, adapt its behaviour or initiate appropriate actions.

SIGNAL PROCESSING

Signal processing must play a substantive role in acquiring, filtering, analysing or interpreting relevant signals and extracting useful information that informs the agent's decisions.

Possible inputs include acoustic, visual, thermal, vibration, seismic, environmental-sensor and remote-sensing data. Applications may include wildlife monitoring, habitat change, environmental threat detection and human-wildlife interaction. These are examples, not assigned problem statements.

No dataset, signal modality, architecture, model or processing pipeline is prescribed. Software prototypes, simulations and hardware-assisted implementations are all acceptable; dedicated hardware is not compulsory.

RESEARCH DIRECTION

A complete product is not expected. Strong submissions will formulate a meaningful technical question, evaluate their approach with appropriate evidence and explain its limitations. A working core with credible results can be stronger than a polished interface without technical depth.



EVALUATION

Grading rubric

Projects will be assessed on technical substance, evidence and research potential. The prototype does not need to be fully complete, but claims must be supported by the work demonstrated.

CRITERION	POINTS	WHAT IS ASSESSED
Research novelty and originality	15	A clear, non-routine contribution or a fresh approach to the problem.
Technical integration and depth	25	Meaningful contribution from both Agentic AI (13 points) and Signal Processing (12 points). Neither should be superficial.
Experimental results and evaluation	20	Use task-appropriate metrics and test conditions; interpret results honestly. Include a baseline where feasible and report compute or energy costs where relevant.
Problem significance and relevance	10	A meaningful problem within the forests and wildlife theme, with clear motivation and sensible scope.
Research potential and scalability	15	A credible next research question, further experiments or potential to generalise the approach.
Prototype implementation and evidence	10	A technically credible working core supported by results, outputs or other inspectable evidence.
Reproducibility and documentation	5	Clear account of methods, data sources, assumptions and relevant setup details.
TOTAL	**100**	

TIE-BREAK Technical Integration and Depth takes priority, followed by Experimental Results and Evaluation, then Research Novelty and Originality.

A NOTE ON INCOMPLETE WORK

A feature may remain unfinished. Teams should distinguish completed work from planned work and report results, limitations and unsuccessful experiments honestly. A live demonstration is not mandatory; implementation evidence and a clear explanation may also be used.

GOOD LUCK

Think clearly. Build thoughtfully. Let the results speak.